Ilyaas Kapadia

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Available for Spring 2027 Co-op

Education & Skills

University of Cincinnati · Bachelor of Science + Master of Engineering, Computer Science — May 2029
Classes: Discrete Math, Computer Systems, Data Structures
Skills: C, C++, Python, Go, Bash, SQL · Linux, Git, Neovim, Codex, Tailscale, OMP, Tern, MCP

Work Experience

Software Engineer Intern · Kernel (YC S25) — May 2026 – Aug 2026
- Owned Programmatic API Key Generation - SDK, MCP, CLI, Dashboard implementations
- Created Kernel's Terraform Provider **(Infrastructure as Code)** from scratch
- Reduced Docker Image Size by 300MB and build time by 50s across images
- Saved **$23,045**/year by optimizing CI/CD pipelines
- Prototyped a web browser onboarding flow via the CLI (importing browser settings)

Sales Consultant · RealNZ — Dec 2023 – Sep 2024
Queenstown, New Zealand
- Closed **800+** sales worth **$250,000+** (Top Sales Consultant, Summer 2024)
- Reduced Whakatipu Vessel data entry time by **11%** with **Python**

Founder & Owner · Shralp Jersey — Oct 2020 – Mar 2023
Mountain Bike (MTB) Jersey on Amazon India
- Identified market opportunity in underserved Direct to Consumer Indian MTB apparel space
- Scaled to **#1 Downhill MTB Jersey Brand** in India through organic traffic and word of mouth

Projects

Reinforcement Learning Simulation Engineering · PufferLib, C, CUDA, MuJoCo — Sep 2026
- Built a self-play CUDA C Slither.io environment (3M SPS) at **ilya.as/slither**
- Built a sim for a 2 DoF self rising robot (3M SPS) With MuJoCo Warp, CUDA port of Box3D **here**
- Built a Brotato RL Env by reverse engineering the game, stripped the GoDot game engine, parallelized physics and contact solving with CUDA (5M SPS). Created a game mod to eval the policy in game (sim2real).

kstack · Kotlin, Android, Tailscale SSH — Sep 2026
- Built an e-ink Android client for driving a fleet of `cli` agents over persistent Tailscale SSH connections, mapping herdr RPCs to a terminal UI for agent control

Bearcat Motorsports EV Battery Management System · C++, Python — Sep 2025
- Built Python **UART** interface to **LTC6813 BMS IC** with stateful file-per-dataset handshake and RX buffer flushing, for real-time battery data logging for **FSAE EV** testing

NES Eval · Python, GitHub Actions — Jan 2026
- Built an eval that gives coding agents 24 hours to write a complete NES emulator with WebAssembly support, graded against Rusty NES.

Reading & Major Influences

- Rich Hickey's Simple Made Easy
- Rich Hickey's Hammock Driven Development
- Efron Licht's Starting Systems Programming Series
- Boot.dev's Data Analysis, Backend, DevOps Paths
- The Algorithm
- The Manga Guide to Microprocessors
- The Goal
- Computer Systems: APP (in-progress)
- Efron Licht's Backend Basics Series
- The Manga Guide to Databases
- Inference Engineering
- Open Data Structures (in-progress)

Leadership

National Level Downhill Mountain Biker — 2019 – 2023
- Assistant Trail Builder at **Largest Mountain Bike Event** in India to date
- Founded a local Mountain Bike community. Built trail network with 6 trails among 2 hills over 2 years from scratch